Fine fragmentation of the high-burnup structure in SCIANTIX

Roadmap and first results

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Context

SCIANTIX & HBS fragmentation modelling

- HBS formation: restructured volume fraction α
- HBS porosity: pore density N_p, mean volume $\bar{V}$, Xe atoms per pore $\bar{n}$ and their variance $\mathrm{Var}(n)$
- HBS fragmentation: which pores break, how much gas they release, fragment size

Rupture per pore class, progressive

- 81 pore classes rebuilt at every step from the state of the porosity model: $(\bar{V}, \bar{n}, \mathrm{Var}(n))$
- a rupture criterion on every class; the broken classes vent their content
- outputs: D = fraction of broken pores (fragment size), G = fraction of the pore gas released, d = mean fragment size
- no parameter fitted on the test shown below

Rupture criterion: Jernkvist 2019 and OperaHPC D5.2

$$\hat{s}\left(p - \frac{2\gamma}{R} - P_h\right) \geq \sigma_R + P_h \iff p \geq P_{cr} = P_h + \frac{2\gamma}{R} + \frac{\sigma_R + P_h}{\hat{s}}$$

Jernkvist 2019 (Olander)

$$P_{cr} = \frac{2\gamma}{R} + \frac{\sigma_{hbs}^{cr}(1-\xi) + P_h}{\xi},$$

$$\sigma_{hbs}^{cr} = 21 \text{ MPa} \quad \hat{s} = \frac{\xi}{1-\xi}$$

OperaHPC D5.2

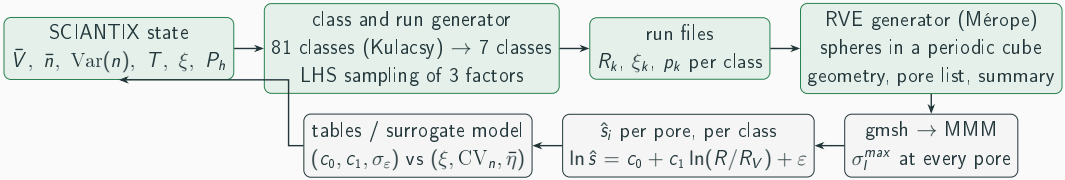
$$\sigma_I^{max}|_i \geq \sigma_R = 275 \text{ MPa},$$

$$\hat{s}_i = \frac{\sigma_I^{max}|_i(1, p_{hyd}) + p_{hyd}}{1 - p_{hyd}}$$

$\hat{s}$ is the stress concentration factor at the pore that will enter in SCIANTIX following micromechanics calculation. Everything else in the criterion comes from the SCIANTIX state.

Scope and chain of tools

Goal: replace the mean-field stress factor $\xi/(1-\xi)$ of Jernkvist by tables computed with MMM on RVEs that reproduce the pore classes that SCIANTIX will see.



green: done; grey: next.

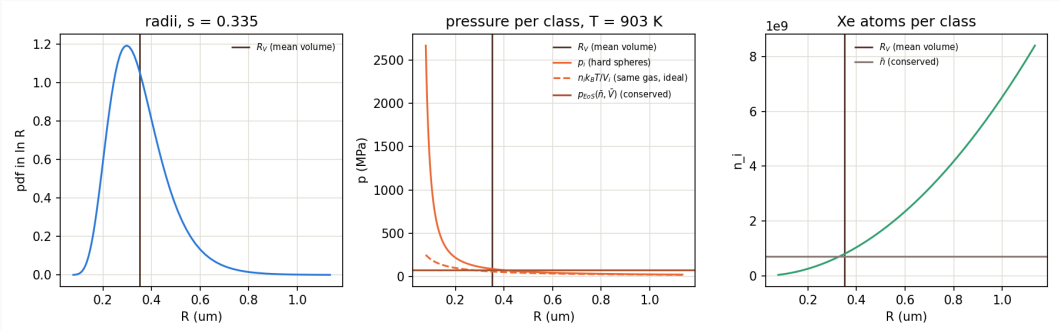
Pore classes from SCIANTIX - Pore population generator I

Porosity model from SCIANTIX: $\bar{V}$ (R_V), $\bar{n}$ = Xe-atoms/pore, $\mathrm{Var}(n)$, N_p, $\xi = N_p \bar{V}$.

log-normal R (K15) : 81 nodes $x_i \in [-4, 4]$, $w_i \propto e^{-x_i^2/2}$, $R_i = R_V\, e^{s x_i} \Big/ \left(\sum_j w_j e^{3 s x_j}\right)^{1/3}$ $\left(\sum_i w_i V_i = \bar{V}\right)$

Xe/pore distribution : $n_i = \bar{n}\, R_i^a \Big/ \sum_j w_j R_j^a$, $a = 2$ (gas density $\propto 1/R$), s from $\sum_i w_i (n_i - \bar{n})^2 = \mathrm{Var}(n)$

pressure distribution : $p_i = \dfrac{n_i k_B T}{V_i} Z(\eta_i)$, $\eta_i = \min\left(\dfrac{n_i v_{Xe}(T)}{V_i}, 0.65\right)$ hard spheres



NFIR, end of base irradiation: $s = 0.33$ (measured about 0.3, K15 0.356), radii 0.08–1.1 μm, p_i from 2.7 GPa to 30 MPa.

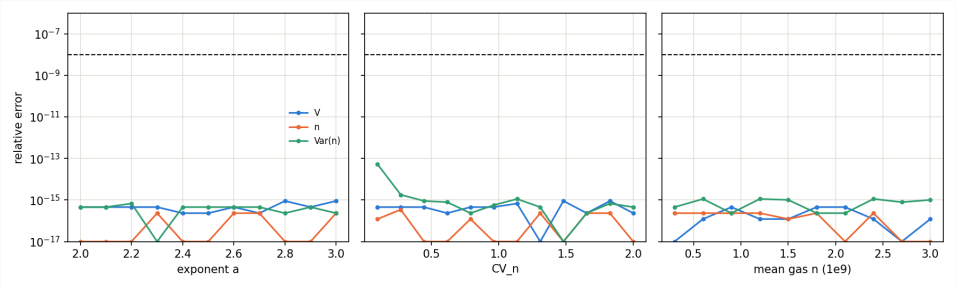
Pore population generator II

$$\sum_i w_i V_i = \bar{V} \qquad \text{exact by construction}$$

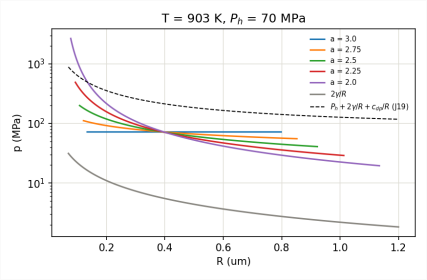
$$\sum_i w_i n_i = \bar{n} \qquad \text{exact by construction}$$

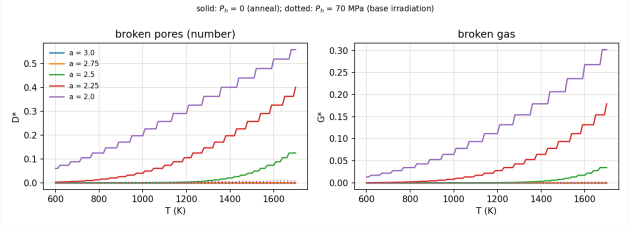
$$\mathrm{CV}_n^2(s, a) = \frac{\sum_i w_i e^{2asx_i}}{\left(\sum_i w_i e^{asx_i}\right)^2} - 1 = \frac{\mathrm{Var}(n)}{\bar{n}^2} \qquad \text{root in } s$$

Mean pressure (pressure of the volume-weighted density): $p_{EoS}(\bar{n}, \bar{V}, T)$ is conserved.



Pore population generator III: Xe atoms/pore





The moments hold for every a; what changes is the width s, the pressure spread and which classes break: small pores first for $a < 3$, while for $a = 3$ uniform pressure.

Reduction of the parameter space and design of the runs

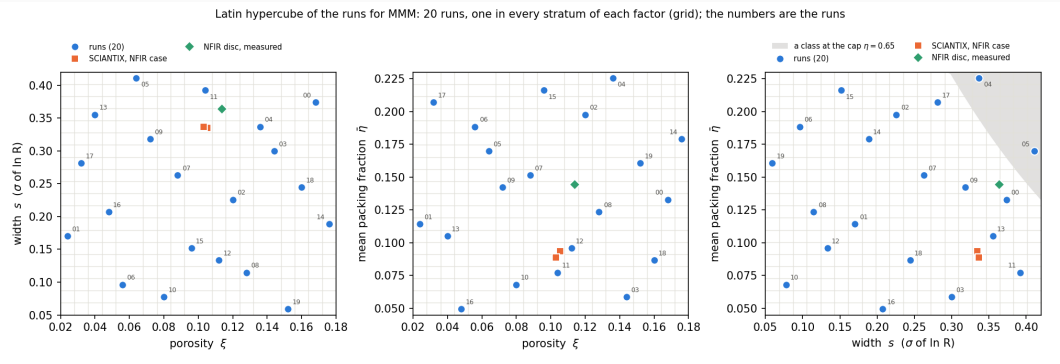
MMM is scale-free in length and linear in the load: R_V and $\bar{p}$ only set the scale. With $p^* = p\, v_{Xe}/k_B T$,

$$p_i^* = \eta_i Z(\eta_i), \qquad \bar{\eta} = \frac{\bar{n}\, v_{Xe}(T)}{\bar{V}}, \qquad \mathrm{CV}_n = \frac{\sqrt{\mathrm{Var}(n)}}{\bar{n}}$$

so in $x_i = R_i/R_V$ the profile $p_i/\bar{p}$ and the weights w_i depend on $(\bar{\eta}, \mathrm{CV}_n, a)$ only.

factor	sets	range
ξ	porosity, pore distances	0.02–0.18
s (from CV_n)	R_k/R_V and ξ_k/ξ	0.05–0.42
$\bar{\eta}$	with s, the profile $p_k/\bar{p}$	0.045–0.23

Latin Hypercube Sampling: 7 pore classes per run (groups the 81 SCIANTIX nodes); Scale: $R_V = 0.73$ μm of the disc and the temperature of the anneal. Ranges from SCIANTIX on the NFIR history and from the measured disc; ξ below 0.02 leaves less than two pores in the smallest class.


Latin hypercube of the runs for MMM: 20 runs, one in every stratum of each factor (grid); the numbers are the runs
runs (20)
SCIANTIX, NFIR case
NFIR disc, measured
width s (σ of ln R)
porosity ξ
mean packing fraction η̄
a class at the cap η = 0.65

Mérope: geometry of a run

For every run file:

1. read the classes, sort by decreasing radius
2. expected pores per class $= \xi_k / V_k \cdot L^3$
3. throw the spheres by RSA in a periodic cube, with a repulsion distance between surfaces
4. each sphere takes the class of the nearest radius, hence its pressure; one phase id per sphere, not meshed
5. write, per run and seed: the pore list (id, x, y, z, R, p), the .geo for gmsh, a summary

Fixed:

cube	30 μm	41 R_V of the disc
repulsion	0.3 μm	one HBS grain, as OperaHPC
mesh	0.15 μm	as OperaHPC
seed	1 per run	

Tests on the RVE: size and mesh

Quantity under test: Z_k, the mean of $\hat{s}$ over the pores of class k in a cube. The proposed tests follow Kanit et al. 2003 (Int. J. Solids Struct. 40, 3647) and the work done in OperaHPC.

Test 1: size of the cube

- **base**: 10 cubes of 30 μm (different seeds), mesh 0.15 μm
- scatter D of Z_k between the 10 cubes: the 30 μm cube is an RVE for class k if $2D/\bar{Z} \leq$ Error %

Test 2: size of the mesh

- a base cube re-meshed at 0.30 μm, the same pores; compared with its 0.15 μm twin
- $\hat{s}$ pore by pore and the mean of every class on the two meshes
- The corser mesh could be tested against the standard one, and also it can be refined within mmm to save computational cost.

NFIR test

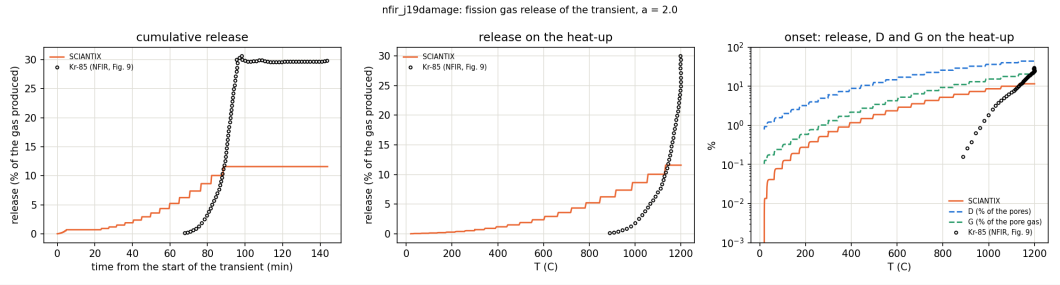
The test (Noirot 2014; OperaHPC D5.2)

- UO_2 disc irradiated in Halden to 103 GWd/t at 560–700 °C, HBS over the whole disc, porosity 11 %, 88 % of the remaining Xe in HBS pores
- measured: pore size distribution before and after, gas density in the pores, temperature history
- anneal: ramp to 300 °C, hold, ramp to 1200 °C at 0.2 °C/s, hold 15 min; Kr-85 release online
- release of about 30 % of the gas produced, in two steps above 880 °C, nothing during the hold; the disc breaks into pieces below 100 μm

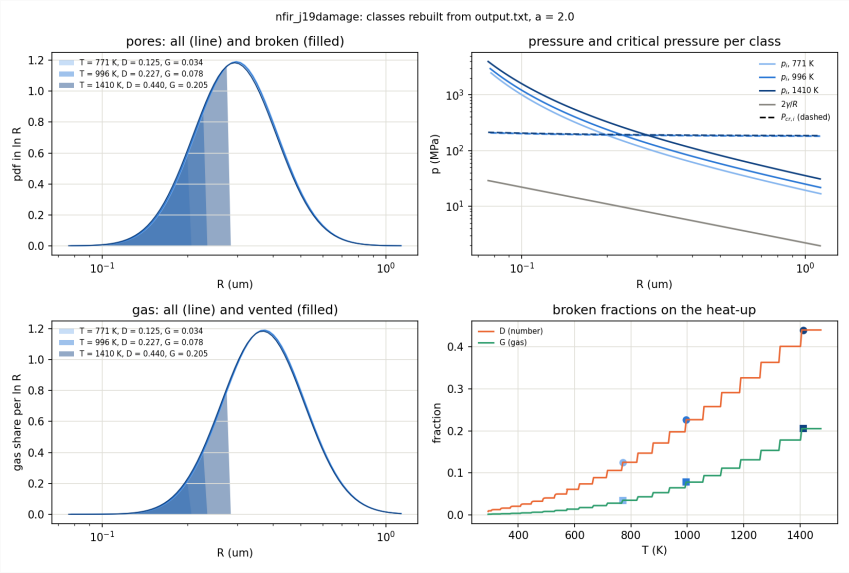
State at the end of the base irradiation from SCIANTIX: $\xi = 0.105$, $R_V = 0.35$ μm, $\bar{n} = 7.2 \times 10^8$ Xe/pore, $\mathrm{CV}_n = 0.75$ (measured disc: $R_V = 0.73$ μm, 1.7 times the gas density)

How SCIANTIX performs on NFIR: release

No parameter fit



How SCIANTIX performs on NFIR: release



No parameter fit

Roadmap

Roadmap

Done

- Draft model implemented in SCIANTIX
- NFIR anneal: release 39 % of the measured, wrong shape; the levers identified
- pore classes determined from the moments, LHS spans for all the factors
- Mérope geometry of the runs; proposed two RVE tests

Next, in order

1. test on RVE size and mesh on the 11 cubes: cost of a cube, format of the output, RVE and mesh verdicts;
2. fill the LHS script with the RVE and mesh verdicts, run Merope for the cases: $\hat{s}_i$ per pore class
3. tables / surrogate $(c_0, c_1, \sigma_\varepsilon)$ against $(\xi, \mathrm{CV}_n, \bar{\eta})$;
4. validation runs outside the LHS
5. tables / surrogate into sciantix;